Quantifying the Relationship Between Audio Characteristics and Music Popularity

Gengyuan Ye, Bohan Yang, Haomin Gu, Yinan Wang

Background

Streams, along with the development of digital streaming media platform, has become a decisive evaluation of commercial successfulness for a song. The popularity of a song is largely dependent on several physical and emotional attributes, such as BPM, danceability, liveness, etc. However, traditional music markets generally base on artists' personal musical instinct and inspiration, lacking of an objective and big data-based perspective to examine public preference. We aim to generate a regression model in order to accurately predict the popularity of songs with their musical characteristics. The model can assist composers composing more preferred music and styles and promote the development of music industry.

Abstract

The observational study investigated the impact of multiple internal musical attributes (bpm, speechiness, liveness, etc.) on streams, a quantitative indicator of music popularity. Sourced from Kaggle, the processed data from "2023 Spotify Most Streamed Songs Dataset", after removing incomplete data and error information, included 805 songs (n=805). Under a 0.1 level of significance, multiple linear regressions are first established, and in order to modify the

nonlinear trend of scatterplot, the investigation involved square and inverse transformation of several variables, completing the optimization of the model. Additionally, the investigation adopts stratified analysis of key, mode and singers to verify the model. The result shows that the optimized model can only explain less than 4% of the dataset, but it opens up this innovative research and leads future study to improve the model by considering more complicated external variables and adopt advanced regression.

Aim

We aim to analyze data from the most popular songs on Spotify in 2023 to investigate how different musical features influence song popularity. We seek to develop a model capable of predicting song popularity to function as reference for composers to produce more preferred songs. We also expect to design tests and analysis in order to verify the accuracy of the models and identify shortcomings and areas that require improvement.

Through this research, we hope to gain a better understanding of which musical features are most likely to appeal to listeners, as well as to identify trends in the development of contemporary popular music and listener preferences.

Methodology

To identify the factors affecting popularity, we use the total number of streams for each song on Spotify as the evaluation criterion and examine several musical

features as explanatory variables, including bpm (tempo), Danceability, Valence (emotional positivity), Energy, Acousticness, Instrumentalness, Liveness, and Speechiness.

We investigate whether there are relationships between these musical features and song popularity through regression analysis by Stata 15. We utilized linear regression, quadratic regression and inverse proportional regression in Stata 15 and we removable the variables with P-value greater than the significant value after regressions to eliminate variables independent with the response variable. Since each song has a specific key and mode, we group songs with the same key and mode into stratums to examine whether the results are consistent across different groups. Furthermore, considering that different artists may have distinct musical styles, we also conduct separate analyses for artists with a large number of songs in the dataset to evaluate the applicability of the model across different artist groups.

Results

Stata 15 employs linear regression among all the control variables (bpm, danceability, valence, etc.) and the response variable (streams) and it generates a rough regression model:

The result of linear regression for all the control variables and the response

streams	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
bpm	-732479.3	663079	-1.10	0.270	-2034069	569110.8
danceability	-3796804	1477236	-2.57	0.010	-6696544	-897065.3
valence	99276.92	934765.6	0.11	0.915	-1735620	1934174
energy	-1655627	1500520	-1.10	0.270	-4601072	1289818
acousticness	-1306733	915605.7	-1.43	0.154	-3104020	490553.9
instrumentalness	-3231560	2116046	-1.53	0.127	-7385249	922129.2
liveness	-2580890	1370921	-1.88	0.060	-5271938	110157.1
speechiness	-4428289	1853265	-2.39	0.017	-8066154	-790424.8
_cons	1.05e+09	1.75e+08	6.00	0.000	7.06e+08	1.39e+09

$$Streams = -732479.3 \times bpm - 3796804 \times danceability + 99276.92 \times valence - 1655627 \times energy - 1306722 \times acousticness - 3231560 \times instrumentalness - 2580890 \times liveness - 4428289 \times peechiness + 1050000000 + \varepsilon$$

The regression model for all the control variables and the response

However, the P-values of some variables are greater than the significant level, 0.1. Therefore, we remove these variables and generate another regression based on the remaining control variables, since we momentarily consider the variables with P-value above 0.1 as independent with the response variable. The improved regression model is generated as:

The result of linear regression for the remaining control variables and the response variable (the significant level is 0.1)

The regression model for the remaining control variables and the response variable (the significant level is 0.1)

streams	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
danceability	-3074946	1280678	-2.40	0.017	-5588828	-561063.6
liveness	-2491018	1358342	-1.83	0.067	-5157348	175312.8
speechiness	-4310884	1838864	-2.34	0.019	-7920446	-701322.3
_cons	7.66e+08	9.21e+07	8.32	0.000	5.86e+08	9.47e+08

$$Streams = -3074946 \times danceability - 2491018 \times liveness - 4310884 \times speechiness + speechiness + 7.66 \times 10^8 + \varepsilon$$

To consider the possible correlation of some control variables with relatively low P-values, we also utilize Stata 15 to generate a regression model including control variables under the significant level of 0.2:

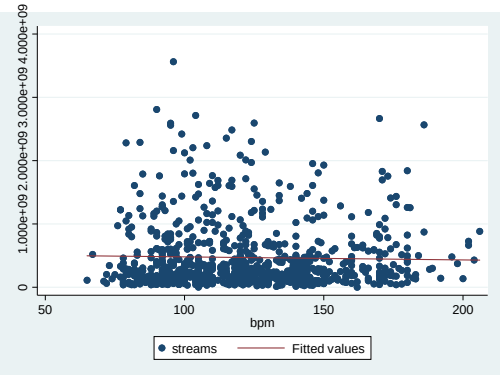
streams	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
danceability	-3558706	1326779	-2.68	0.007	-6163091	-954321.7
acousticness	-691930.3	744151.1	-0.93	0.353	-2152652	768791.9
instrumentalness	-3158663	2099761	-1.50	0.133	-7280362	963035.6
liveness	-2699393	1363098	-1.98	0.048	-5375069	-23717.91
speechiness	-4455744	1843608	-2.42	0.016	-8074630	-836857.3
_cons	8.28e+08	1.03e+08	8.08	0.000	6.27e+08	1.03e+09

The result of linear regression for the remaining control variables and the response variable (the significant level is 0.2)

The regression model for the remaining control variables and the response variable (the significant level is 0.2)

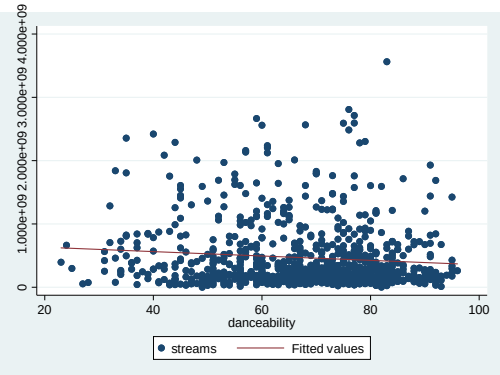
$$\begin{aligned} Streams &= -3074946.1 \times danceability - 1306733.4 \\ &\times acousticness - 3231560.2 \times instrumentalness \\ &- 249101.84 \times livevess - 4431088.4 \times speechiness \\ &+ 766000000 + \varepsilon \end{aligned}$$

However, it doesn't mean that variables with a large P-value are literally independent with the response variable, since linear regression cannot reveal the correlation between control variables after squared or inversed and the response variable. Therefore, according to the distributions and shapes of data in the scattered plots, we predict the regression model of each control variables as either a parabola or a hyperbola. For instance, the distribution of bpm is concentrated in the middle cluster, so we predict it as a parabola. The scattered plot of each control variables and streams are showed on the next page.



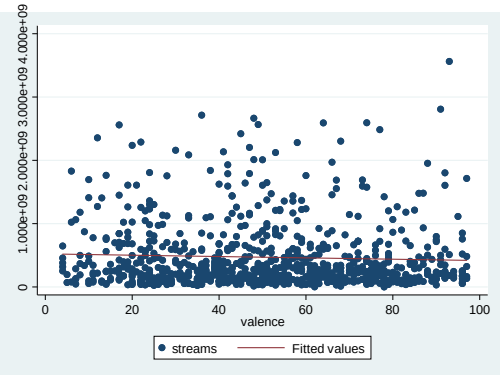
bpm

Predicted as



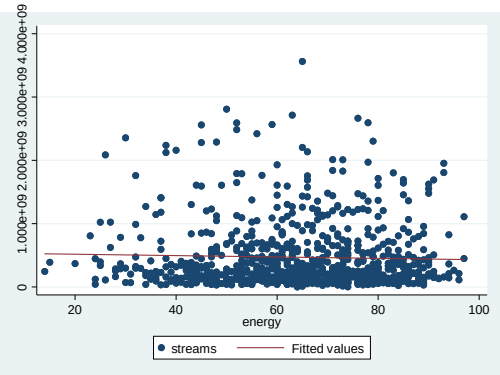
danceability

Predicted as



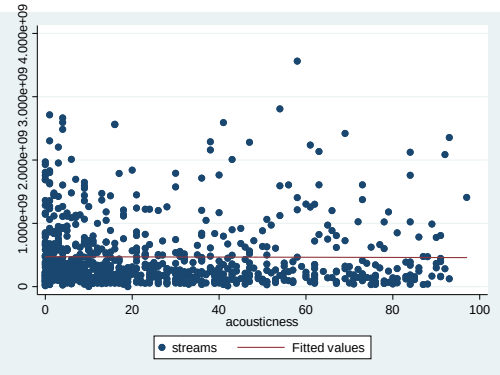
valence

Predicted as



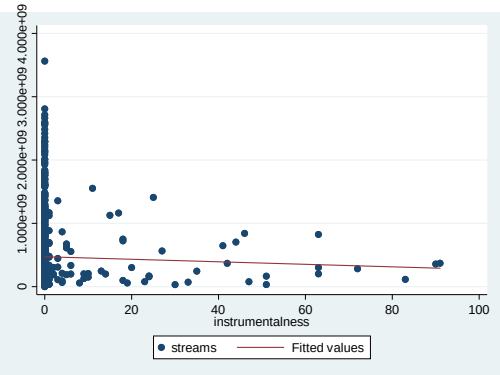
energy

Predicted as



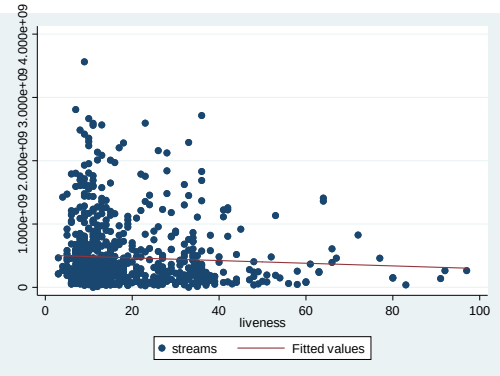
acousticness

Predicted as



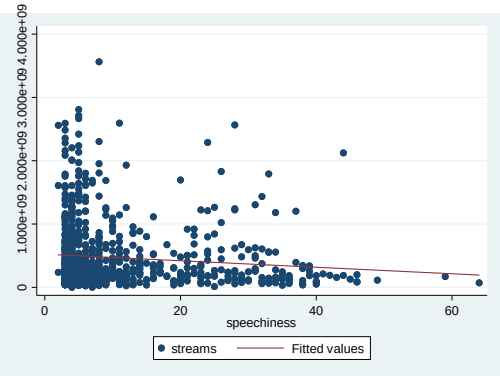
instrumentalness

Predicted as



liveness

Predicted as



speechiness

Predicted as

Then, we calculated the square values of bpm, danceability, valence and energy. We also calculated the inverse value of acousticness, instrumentalness, liveness and speechiness. After we employed quadratic regression or inverse proportion regression for every control variable, we choose the appropriate model according to the smaller P-value between the two regressions. For example, since bpm has a P-value of 0.464 when receiving a linear regression and has a P-value of 0.051 when receiving a quadratic regression, we will suppose that the quadratic regression model is more appropriate for the variable bpm. However, there are too many “0” in the instrumentalness list, indicating we are not able to regress it with an inverse proportional model.

Determination for which regression model to use for each control variable			
variable	P-value in the first regression	P-value in the second regression	Chosesn model
bpm	0.464	0.051	quadratic
danceability	0.006	0.749	linear
valence	0.179	0.358	linear
energy	0.343	0.812	linear
acousticness	0.844	0.132	inverse
instrumentalness	0.344	/	linear
liveness	0.123	0.262	linear
speechiness	0.005	0.747	linear

The comparison between P-value from the two regression is used for all variables to determine which regression model to apply for each variable

Hence, we generated a new regression on all the variables based on the results above, and we retain the variables that have P-values smaller than 0.1. The final regression that leads to our optimal model is showed below:

```
. regress streams bpm bpmsq danceability N liveness speechiness
```

Source	SS	df	MS	Number of obs	=	755
				F(6, 748)	=	4.27
Model	6.7880e+18	6	1.1313e+18	Prob > F	=	0.0003
Residual	1.9828e+20	748	2.6508e+17	R-squared	=	0.0331
				Adj R-squared	=	0.0253
Total	2.0506e+20	754	2.7197e+17	Root MSE	=	5.1e+08

streams	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
bpm	-1.05e+07	5537638	-1.90	0.058	-2.14e+07	351223
bpmsq	38415.91	21515.69	1.79	0.075	-3822.417	80654.23
danceability	-2595506	1411964	-1.84	0.066	-5367389	176377.4
N	1.99e+08	7.75e+07	2.57	0.010	4.71e+07	3.51e+08
liveness	-2516920	1379948	-1.82	0.069	-5225953	192112.3
speechiness	-4611638	1902919	-2.42	0.016	-8347335	-875941.7
_cons	1.39e+09	3.31e+08	4.18	0.000	7.35e+08	2.04e+09

The result of regression for the optimal model
(N indicates 1/acousticness)

The form of the optimal model

$$Streams = -1.05 \times 10^7 \times bpm + 38415.91 \times bpm^2 - 2595506 \times danceability + 1.99 \times \frac{10^8}{acousticness} - 2516920 \times liveness - 4611638 \times speechiness + 1.39 \times 10^9 + \varepsilon$$

This is the best model we can generate in limited time, all the P-values are kept in very timid magnitude, but the R-squared value is only 0.0331, which means that our optimal model cannot accurately predict the streams of each music. We hypothesized that the reason why the optimal model is so weak is because other non-quantitative variables are also influential to the number of streams. Thus, we also pay effort in stratified analysis to prove our hypothesis. After

the exclusion of dataset with less than 30 data, we have 13 groups of datasets with different combinations of keys (from A to G#) and modes (major and minor). We utilized computer tools to perform SRS on the 13 datasets and pick out the datasets named (key + mode): E Minor and F# Minor. We regress the data using the two linear models and the optimal regression model. It means that we will regress the streams of the data with the variables contained in each model. According to the R-square value and the constants of each variable in the regression, we can learn whether the model can accurately explain data in different cases.

The result of the regression with the variables contained in the linear regression model with the significant level of 0.2 in E Minor group

```
. regress streams danceability liveness speechiness instrumentalness acousticness
```

Source	SS	df	MS		
Model	2.7812e+18	5	5.5623e+17	Number of obs	= 44
Residual	9.9999e+18	38	2.6316e+17	F(5, 38)	= 2.11
				Prob > F	= 0.0848
Total	1.2781e+19	43	2.9723e+17	R-squared	= 0.2176
				Adj R-squared	= 0.1147
				Root MSE	= 5.1e+08

streams	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
danceability	-1.37e+07	6515622	-2.10	0.043	-2.69e+07	-485995.7
liveness	-6634320	5344731	-1.24	0.222	-1.75e+07	4185524
speechiness	-808668.2	9117498	-0.09	0.930	-1.93e+07	1.76e+07
instrumentalness	-8423174	6548933	-1.29	0.206	-2.17e+07	4834448
acousticness	4497080	3265438	1.38	0.177	-2113455	1.11e+07
_cons	1.45e+09	4.83e+08	3.01	0.005	4.75e+08	2.43e+09

The result of the regression with the variables contained in the linear regression model with the significant level of 0.1 in E Minor group

```
. regress streams danceability liveness speechiness
```

Source	SS	df	MS	Number of obs	=	44
				F(3, 40)	=	2.56
Model	2.0615e+18	3	6.8718e+17	Prob > F	=	0.0681
Residual	1.0720e+19	40	2.6799e+17	R-squared	=	0.1613
				Adj R-squared	=	0.0984
Total	1.2781e+19	43	2.9723e+17	Root MSE	=	5.2e+08

streams	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
danceability	-1.47e+07	6347864	-2.32	0.026	-2.76e+07	-1898765
liveness	-7552672	5275014	-1.43	0.160	-1.82e+07	3108529
speechiness	996430.4	9090971	0.11	0.913	-1.74e+07	1.94e+07
_cons	1.61e+09	4.35e+08	3.70	0.001	7.33e+08	2.49e+09

```
. regress streams bpm bpmsq N danceability liveness speechiness
```

Source	SS	df	MS	Number of obs	=	42
				F(6, 35)	=	1.43
Model	2.4911e+18	6	4.1519e+17	Prob > F	=	0.2309
Residual	1.0158e+19	35	2.9022e+17	R-squared	=	0.1969
				Adj R-squared	=	0.0593
Total	1.2649e+19	41	3.0851e+17	Root MSE	=	5.4e+08

streams	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
bpm	2.18e+07	2.48e+07	0.88	0.385	-2.85e+07	7.21e+07
bpmsq	-76221.19	96321.68	-0.79	0.434	-271764.6	119322.2
N	-2.76e+08	3.99e+08	-0.69	0.494	-1.09e+09	5.35e+08
danceability	-1.53e+07	6811470	-2.25	0.031	-2.92e+07	-1497606
liveness	-8036200	5740929	-1.40	0.170	-1.97e+07	3618506
speechiness	2045285	9986645	0.20	0.839	-1.82e+07	2.23e+07
_cons	2.34e+08	1.54e+09	0.15	0.880	-2.89e+09	3.36e+09

The result of the regression with the variables contained in the optimal regression model in E Minor group

The result of the regression with the variables contained in the optimal regression model in F# Minor group

```
. regress streams danceability acousticness instrumentalness liveness speechiness
```

Source	SS	df	MS	Number of obs	=	36
				F(5, 30)	=	2.03
Model	1.5658e+18	5	3.1317e+17	Prob > F	=	0.1031
Residual	4.6330e+18	30	1.5443e+17	R-squared	=	0.2526
				Adj R-squared	=	0.1280
Total	6.1988e+18	35	1.7711e+17	Root MSE	=	3.9e+08

streams	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
danceability	3061606	5556077	0.55	0.586	-8285418	1.44e+07
acousticness	-7065070	3137906	-2.25	0.032	-1.35e+07	-656609.8
instrumentalness	-2.06e+07	4.51e+07	-0.46	0.650	-1.13e+08	7.15e+07
liveness	-1.06e+07	5150343	-2.05	0.049	-2.11e+07	-56018.93
speechiness	2335146	6953869	0.34	0.739	-1.19e+07	1.65e+07
_cons	5.95e+08	4.02e+08	1.48	0.149	-2.26e+08	1.42e+09

```
. regress streams danceability liveness speechiness
```

Source	SS	df	MS	Number of obs	=	36
				F(3, 32)	=	1.10
Model	5.7883e+17	3	1.9294e+17	Prob > F	=	0.3640
Residual	5.6200e+18	32	1.7563e+17	R-squared	=	0.0934
				Adj R-squared	=	0.0084
Total	6.1988e+18	35	1.7711e+17	Root MSE	=	4.2e+08

streams	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
danceability	4179120	5725115	0.73	0.471	-7482557	1.58e+07
liveness	-9169710	5451712	-1.68	0.102	-2.03e+07	1935064
speechiness	3614360	7387443	0.49	0.628	-1.14e+07	1.87e+07
_cons	2.83e+08	3.98e+08	0.71	0.482	-5.28e+08	1.09e+09

The result of the regression with the variables contained in the linear regression model with the significant level of 0.1 in F# Minor group

The result of the regression with the variables contained in the optimal regression model in F# Minor group

```
. regress streams bpm bpmsq N danceability liveness speechiness
```

Source	SS	df	MS	Number of obs	=	36
				F(6, 29)	=	6.59
Model	3.5752e+18	6	5.9587e+17	Prob > F	=	0.0002
Residual	2.6236e+18	29	9.0470e+16	R-squared	=	0.5768
				Adj R-squared	=	0.4892
Total	6.1988e+18	35	1.7711e+17	Root MSE	=	3.0e+08

streams	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
bpm	-5.04e+07	1.94e+07	-2.59	0.015	-9.01e+07	-1.06e+07
bpmsq	182522.2	75414.77	2.42	0.022	28281.7	336762.7
N	1.05e+09	2.20e+08	4.77	0.000	6.00e+08	1.50e+09
danceability	8448724	4375416	1.93	0.063	-500005.2	1.74e+07
liveness	-8289359	3973301	-2.09	0.046	-1.64e+07	-163046.4
speechiness	4563978	5454955	0.84	0.410	-6592656	1.57e+07
_cons	3.09e+09	1.15e+09	2.68	0.012	7.33e+08	5.44e+09

Additionally, we discovered that the songs of Taylor Swift and The Weeknd contained in the list are both more than 30 ones. Hence, we performed a stratified test on songs of each singer, too.

The result of the regression with the variables contained in the linear regression model with the significant level of 0.2 among Taylor Swift's songs

```
. regress streams danceability_ acousticness_ instrumentalness_ liveness_ speechiness_
```

Source	SS	df	MS	Number of obs	=	38
				F(5, 32)	=	0.30
Model	1.9013e+17	5	3.8026e+16	Prob > F	=	0.9098
Residual	4.0693e+18	32	1.2716e+17	R-squared	=	0.0446
				Adj R-squared	=	-0.1046
Total	4.2594e+18	37	1.1512e+17	Root MSE	=	3.6e+08

streams	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
danceability_	2346897	6194679	0.38	0.707	-1.03e+07	1.50e+07
acousticness_	-1597142	2091905	-0.76	0.451	-5858214	2663930
instrumentalness_	-4362055	1.72e+07	-0.25	0.802	-3.95e+07	3.08e+07
liveness_	-382932.4	5192071	-0.07	0.942	-1.10e+07	1.02e+07
speechiness_	2222578	9478821	0.23	0.816	-1.71e+07	2.15e+07
_cons	2.88e+08	3.91e+08	0.74	0.467	-5.09e+08	1.08e+09

```
. regress streams danceability_ liveness_ speechiness_
```

Source	SS	df	MS	Number of obs	=	38
				F(3, 34)	=	0.25
Model	9.1724e+16	3	3.0575e+16	Prob > F	=	0.8612
Residual	4.1677e+18	34	1.2258e+17	R-squared	=	0.0215
				Adj R-squared	=	-0.0648
Total	4.2594e+18	37	1.1512e+17	Root MSE	=	3.5e+08

streams	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
danceability_	3975632	5800598	0.69	0.498	-7812602	1.58e+07
liveness_	255893.7	5045658	0.05	0.960	-9998118	1.05e+07
speechiness_	2321446	9305647	0.25	0.805	-1.66e+07	2.12e+07
_cons	1.26e+08	3.38e+08	0.37	0.712	-5.62e+08	8.14e+08

The result of the regression with the variables contained in the linear regression model with the significant level of 0.1 among Taylor Swift's songs

The result of the regression with the variables contained in the optimal regression model among Taylor Swift's songs

The result of the regression with the variables contained in the linear regression model with the significant level of 0.2 among The Weeknd's songs

```
. regress streams bpm bpmsq N danceability_ liveness_ speechiness_
```

Source	SS	df	MS	Number of obs	=	35
				F(6, 28)	=	1.25
Model	8.1484e+17	6	1.3581e+17	Prob > F	=	0.3102
Residual	3.0341e+18	28	1.0836e+17	R-squared	=	0.2117
				Adj R-squared	=	0.0428
Total	3.8489e+18	34	1.1320e+17	Root MSE	=	3.3e+08

streams	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
bpm	-2.43e+07	1.57e+07	-1.54	0.134	-5.64e+07	7919079
bpmsq	107415.3	59901.87	1.79	0.084	-15288.1	230118.8
N	-2.00e+08	2.57e+08	-0.78	0.443	-7.26e+08	3.26e+08
danceability_	1.47e+07	7647056	1.92	0.066	-1013781	3.03e+07
liveness_	1090457	4827812	0.23	0.823	-8798868	1.10e+07
speechiness_	-9048115	1.01e+07	-0.90	0.377	-2.97e+07	1.16e+07
_cons	8.36e+08	1.04e+09	0.80	0.430	-1.30e+09	2.97e+09

```
. regress streams danceability_ liveness_ acousticness_ speechiness_ instrumentalness_
```

Source	SS	df	MS		
Model	1.8815e+18	5	3.7630e+17	Number of obs	= 33
Residual	1.0991e+19	27	4.0708e+17	F(5, 27)	= 0.92
				Prob > F	= 0.4805
				R-squared	= 0.1462
				Adj R-squared	= -0.0120
Total	1.2873e+19	32	4.0227e+17	Root MSE	= 6.4e+08

streams	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
danceability_	-1.01e+07	8876137	-1.13	0.267	-2.83e+07	8151389
liveness_	-5618225	7867732	-0.71	0.481	-2.18e+07	1.05e+07
acousticness_	-1.32e+07	7580525	-1.74	0.093	-2.87e+07	2361825
speechiness_	-1594366	1.31e+07	-0.12	0.904	-2.85e+07	2.53e+07
instrumentalness_	644801.9	2.57e+07	0.03	0.980	-5.20e+07	5.33e+07
_cons	1.58e+09	6.81e+08	2.32	0.028	1.85e+08	2.98e+09

The result of the regression with the variables contained in the linear regression model with the significant level of 0.1 among The Weeknd's songs

```
. regress streams danceability_ liveness_ speechiness_
```

Source	SS	df	MS	Number of obs	=	37
				F(3, 33)	=	0.46
Model	9.2299e+17	3	3.0766e+17	Prob > F	=	0.7100
Residual	2.1927e+19	33	6.6446e+17	R-squared	=	0.0404
				Adj R-squared	=	-0.0468
Total	2.2850e+19	36	6.3473e+17	Root MSE	=	8.2e+08

streams	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
danceability_	-5022067	9077707	-0.55	0.584	-2.35e+07	1.34e+07
liveness_	-1.05e+07	9403180	-1.12	0.270	-2.97e+07	8589382
speechiness_	-2552566	1.62e+07	-0.16	0.876	-3.55e+07	3.04e+07
_cons	1.18e+09	6.37e+08	1.85	0.073	-1.16e+08	2.48e+09

```
. regress streams bpm bpmsq N danceability_ liveness_ speechiness_
```

Source	SS	df	MS	Number of obs	=	33
				F(6, 26)	=	3.91
Model	6.1063e+18	6	1.0177e+18	Prob > F	=	0.0065
Residual	6.7663e+18	26	2.6024e+17	R-squared	=	0.4744
				Adj R-squared	=	0.3531
Total	1.2873e+19	32	4.0227e+17	Root MSE	=	5.1e+08

streams	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
bpm	-4.93e+07	2.78e+07	-1.77	0.088	-1.07e+08	7859768
bpmsq	273046.7	112868.4	2.42	0.023	41042.46	505051
N	-2.79e+08	4.29e+08	-0.65	0.521	-1.16e+09	6.03e+08
danceability_	-849079	6606500	-0.13	0.899	-1.44e+07	1.27e+07
liveness_	-5402730	6502468	-0.83	0.414	-1.88e+07	7963284
speechiness_	-2.77e+07	1.21e+07	-2.29	0.030	-5.25e+07	-2865261
_cons	2.97e+09	1.75e+09	1.70	0.101	-6.25e+08	6.57e+09

It is obvious that the optimal model can explain the dataset much better than the other

The result of the regression with the variables contained in the optimal regression model among The Weeknd's songs

two in the four cases, while it still needs to be improved since it can only explain data relatively accurately in only a couple of occasions. This may due to two facts: We ignored many other complex factors, such as the popularity of the singer, the tones and modes in songs and the year of establishing the song, etc.; Moreover, there may be more complex models that can accurately reflect the relationship between the control variables and the response variable. However, due to limited time, we can only perform tests on eight variables and only predict the models as simply linear, simply quadratic and simply inverse proportional.

Importance and Innovations

the study is an original study. We were unable to find previous projects that explored this exact topic using the same dataset and methods. This shows that this research performs a “zero to one” step. It provides a starting point for future studies on music popularity.

we successfully identified the general trends between different musical variables and popularity. Although these trends can’t be 100% sure that a song will become popular, they can still provide useful information for musicians. Understanding these patterns may help them make decisions when composing and producing music.

Limitations and imperfections

the data only contains popular songs from 2023, so it can't represent all types of music. Moreover, many important factors affecting popularity were not included in our analysis. In the future research, we should use a lager dataset from all years and platforms, include more variables and apply more advanced statistical or machine learning methods to improve prediction accuracy.

Conclusion

In this project, we studied how different musical features affect a song's popularity using Spotify's 2023 most-streamed songs. Our results show that some musical features are related to popularity. Variables such as danceability, liveness, and speechiness showed stronger relationships with the number of streams than other variables. Although the optimized model performed better than the original linear model, the stratified analysis showed that it did not work equally well for every group of songs classified by mode, key and singers' popularity. In the future, we expect to continue the study by generating a more accurate model considering the impact of more unconsidered variables. Moreover, we will try more complex and advanced models on each variable to ensure the accuracy of the model. We seek to establish our model in the future and make contributions to the music industry.

Bibliography

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